

Department of Electrical, Electronic and Computer Engineering

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Design and Implementation of a Secure Windows Server 2019 Infrastructure for Enterprise Network Services

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by

Group 1

Group project report submitted in partial fulfilment of the requirements for the degree

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in the Faculty of Engineering and the Built Environment

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AI Declaration

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- **LaTeX formatting assistance** (structure and code improvements).

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Abstract

This report presents the installation, configuration, and testing of a Windows Server 2019 environment developed to meet the network and security requirements of CPUT. The project was implemented using the hypervisor utilised during Semester 1 and included the deployment of several key server roles and services.

The implementation involved configuring Active Directory Domain Services (AD DS) for centralized user and resource management, Domain Name System (DNS) for name resolution, and Dynamic Host Configuration Protocol (DHCP) for automatic IP address allocation. In addition, Group Policy settings were applied to improve the security of the environment by restricting Command Prompt access, disabling removable storage devices, preventing unauthorized software installations, and enforcing stronger password policies.

Remote Access Services and Windows Certificate Services were also configured to provide secure connectivity and authentication within the network environment. Each service was tested to verify that it functioned correctly and met the specified requirements. Screenshots of all installation, configuration, and testing procedures were captured and included as evidence of successful implementation.

The results demonstrate how Windows Server 2019 can be used to create a secure and manageable network infrastructure that supports organisational requirements while following recommended security practices.

Keywords: CPUT, Windows Server 2019, Active Directory, DNS, DHCP, Group Policy, Remote Access, Certificate Services, Network Administration, Security Policies, Server Configuration

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1. Introduction

As organisations continue to grow, the need for a reliable and secure network infrastructure becomes increasingly important. CPUT requires an IT environment that can support its expanding operations while ensuring efficient management of users, devices, and network resources. To meet these requirements, a Windows Server 2019 environment was implemented using the hypervisor used during Semester 1.

This project focuses on the installation, configuration, and testing of key Windows Server services, including Active Directory Domain Services (AD DS), DNS, DHCP, Group Policy, Remote Access, and Windows Certificate Services. Screenshots were captured throughout the process to provide evidence of the installation and configuration steps performed.

1.1 Background

Windows Server 2019 is widely used in organisations to manage network resources, user accounts, security policies, and shared services from a central location. Services such as Active Directory, DNS, and DHCP help administrators manage users and devices more efficiently while ensuring that network communication functions correctly.

Security is also an important part of network administration. For this reason, several security policies were implemented in this project. These include disabling Command Prompt access, preventing the use of removable storage devices, restricting software installations, enforcing a minimum password length of ten characters, and requiring users to change their passwords regularly. These measures help reduce security risks and improve the overall protection of the network environment.

1.2 Objectives

The objectives of this report are:

- To install and configure Windows Server 2019.
- To configure and test Active Directory Domain Services (AD DS).

- To configure and test DHCP services.
- To configure and test DNS services.
- To implement Group Policy settings that improve system security.
- To configure Remote Access services.
- To install and configure Windows Certificate Services.
- To verify that all configured services are functioning correctly.
- To document all installation, configuration, and testing procedures using screenshots.

1.3 Report Structure

The remainder of this report is organised as follows:

- Chapter 2 covers the installation and initial configuration of Windows Server 2019.
- Chapter 3 discusses the configuration and testing of Active Directory Domain Services (AD DS).
- Chapter 4 explains the configuration and testing of DHCP services.
- Chapter 5 presents the configuration and testing of DNS services.
- Chapter 6 describes the implementation of Group Policy settings.
- Chapter 7 discusses the configuration of Remote Access services.
- Chapter 8 explains the installation and configuration of Windows Certificate Services.
- Chapter 9 presents the testing and verification of the configured services.
- Chapter 10 provides the conclusion of the project.
- The final section contains the references used in this report.

2. Installation and Initial Configuration of Windows Server 2019

2.1 Introduction

This chapter describes the installation and initial configuration of Windows Server 2019 within the virtualised environment used during Semester 1. The purpose of this stage was to establish a functioning server that could support the deployment of the network services required for the project.

The installation process involved preparing the virtual machine using VirtualBox, installing the operating system, and completing the initial server configuration. These steps provided the foundation for the subsequent configuration of Active Directory Domain Services (AD DS), DNS, DHCP, Group Policy, Remote Access, and Windows Certificate Services.

2.2 Virtual Machine Preparation

Before installing Windows Server 2019, a virtual machine was prepared using VirtualBox. Virtualisation allows multiple operating systems to run on a single physical computer while maintaining separate virtual environments.

The virtual machine was configured with the necessary resources to support the operating system and the planned server roles. These resources included virtual memory, processor allocation, virtual storage, and a network adapter. The Windows Server 2019 installation media was then attached to the virtual machine, allowing it to boot from the installation image.

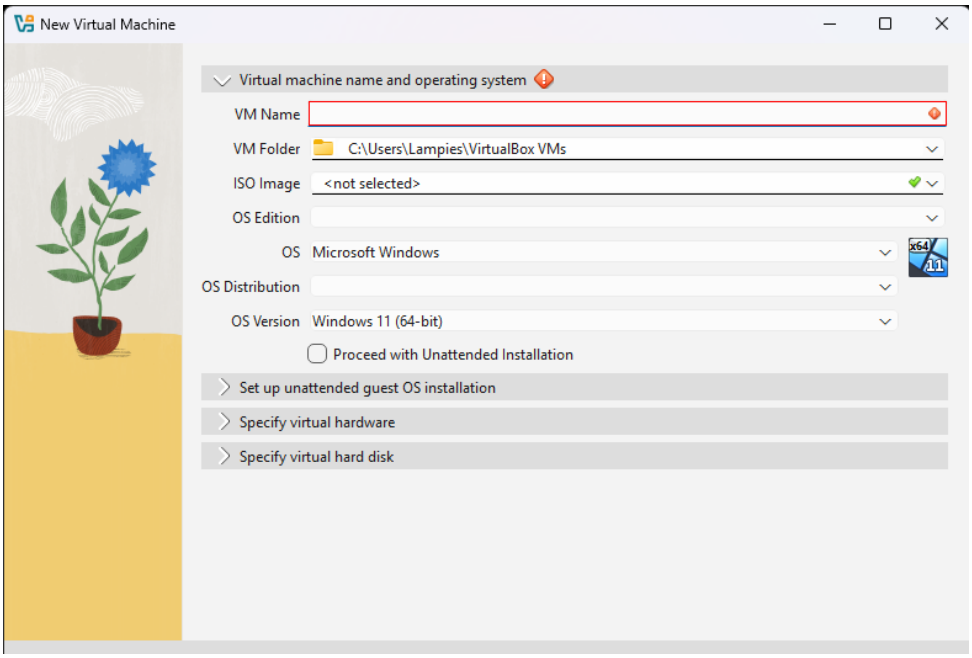


Figure 2.1: Initial Virtual Machine Configuration.

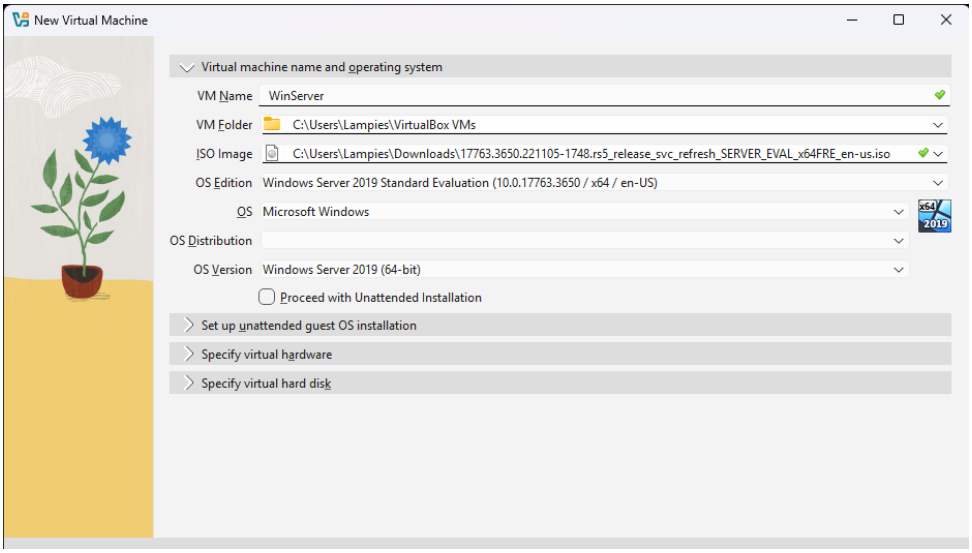


Figure 2.2: Virtual Machine Configuration Step 2.

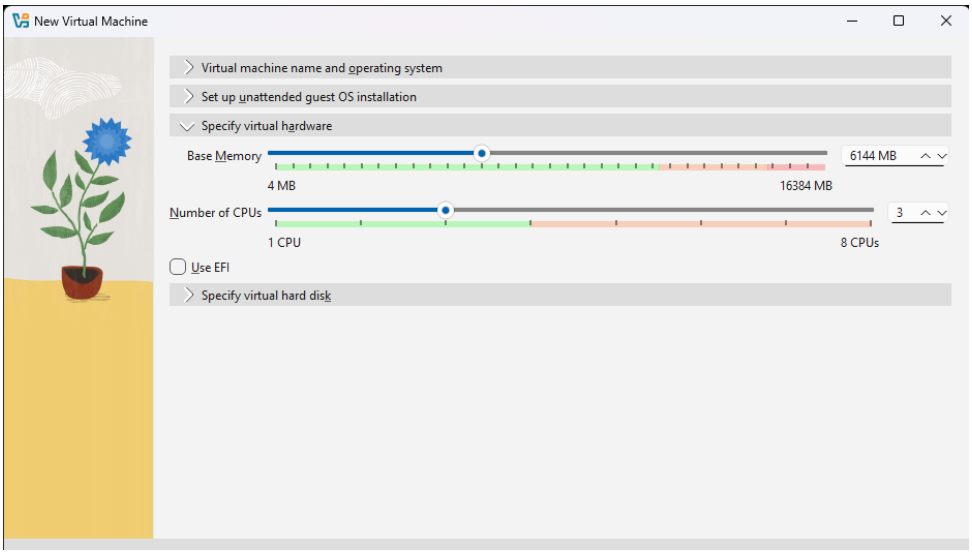


Figure 2.3: Virtual Machine Configuration Step 3.

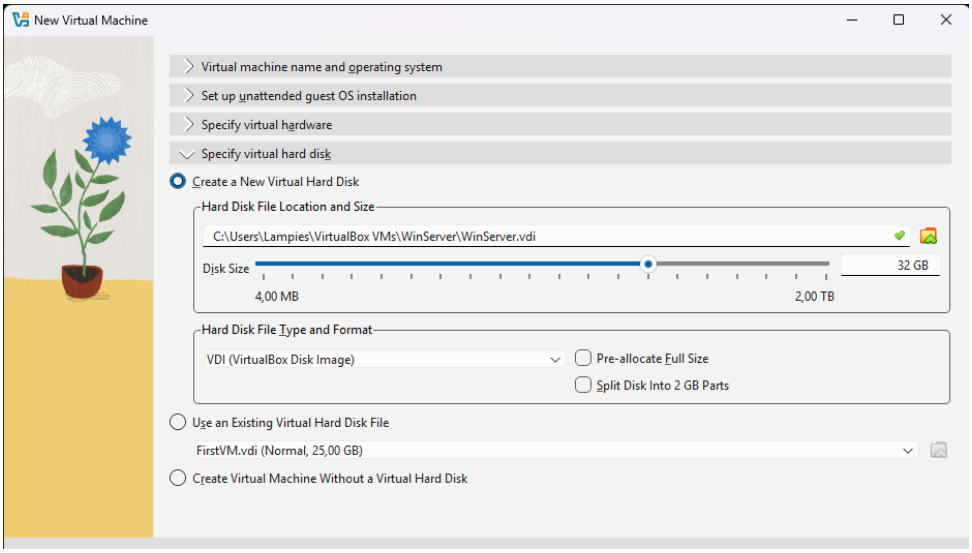


Figure 2.4: Virtual Machine Configuration Step 4.

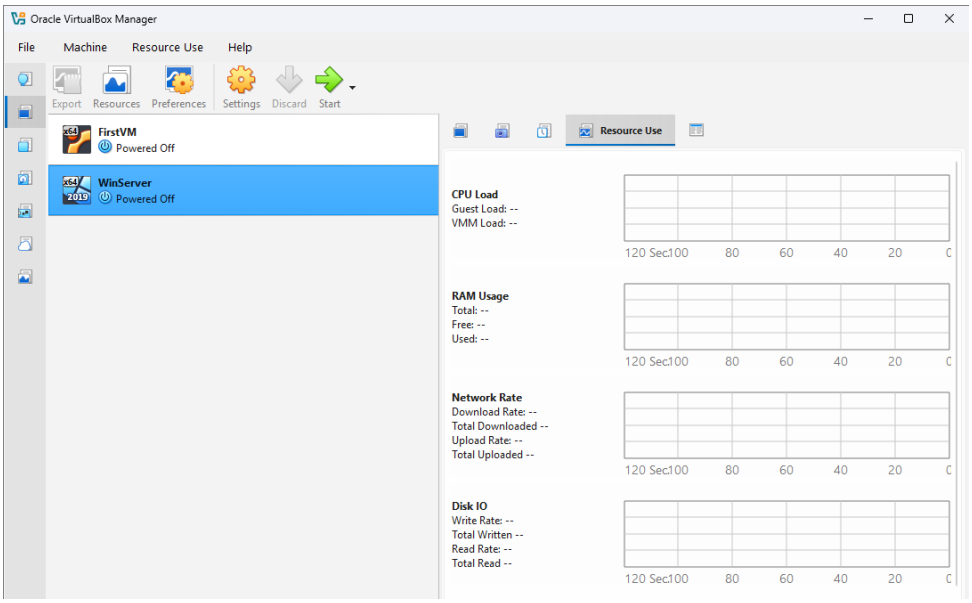


Figure 2.5: Virtual Machine Configuration Step 5.

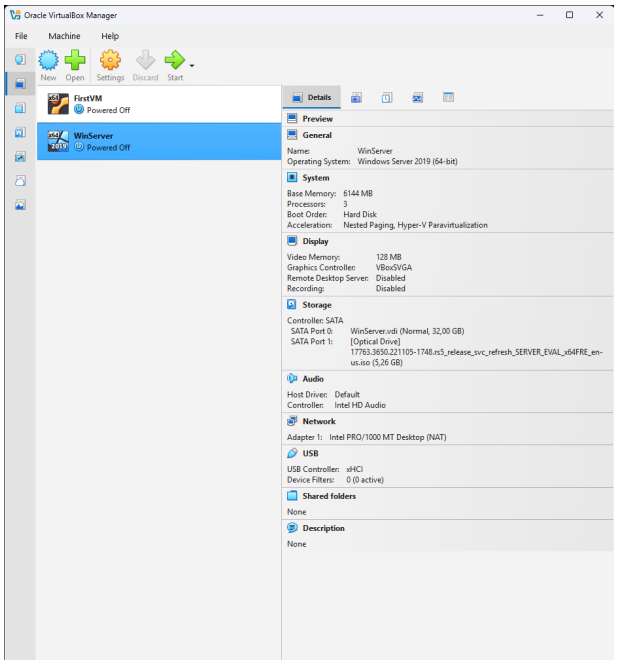


Figure 2.6: Final Virtual Machine Configuration Step.

2.3 Installation of Windows Server 2019

Once the virtual machine had been configured, it was started and the Windows Server 2019 installation process was initiated. The installation wizard was used to select the appropriate language, time and currency format, and keyboard layout.

The required Windows Server 2019 edition was then selected. The installation proceeded by accepting the applicable licence terms, selecting the installation type,

and choosing the virtual disk on which the operating system would be installed.

After the installation files had been copied and the required setup operations completed, the virtual machine restarted. The initial administrator password was configured, allowing access to the server using the local Administrator account.

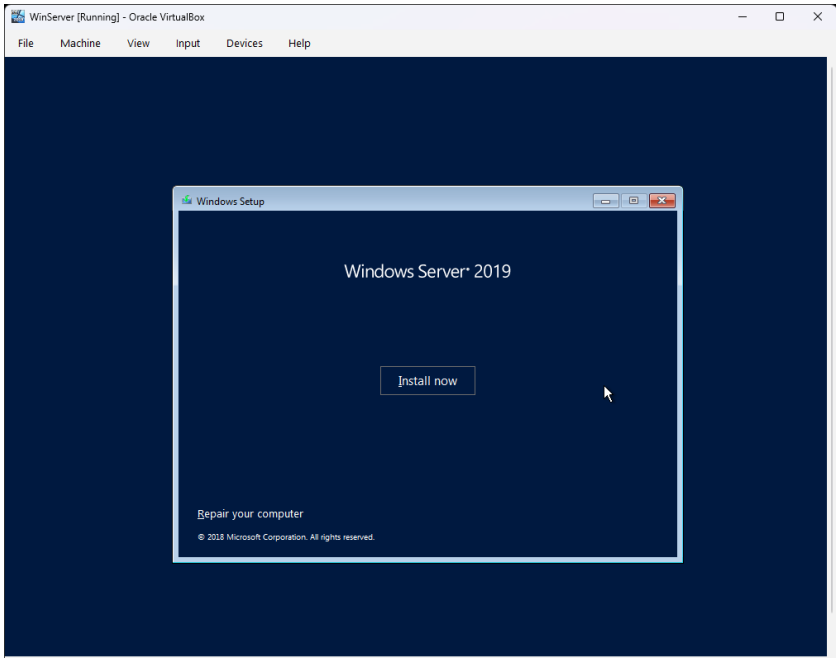


Figure 2.7: Installation Screen 1.

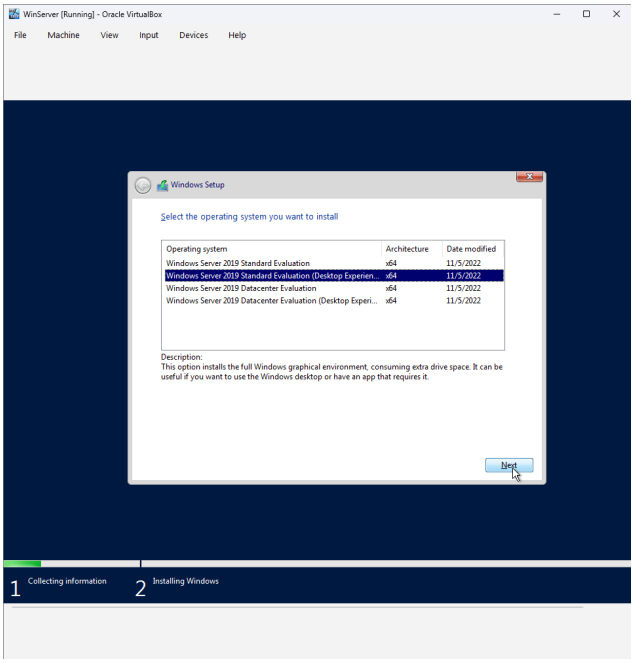


Figure 2.8: Installation Screen 2.

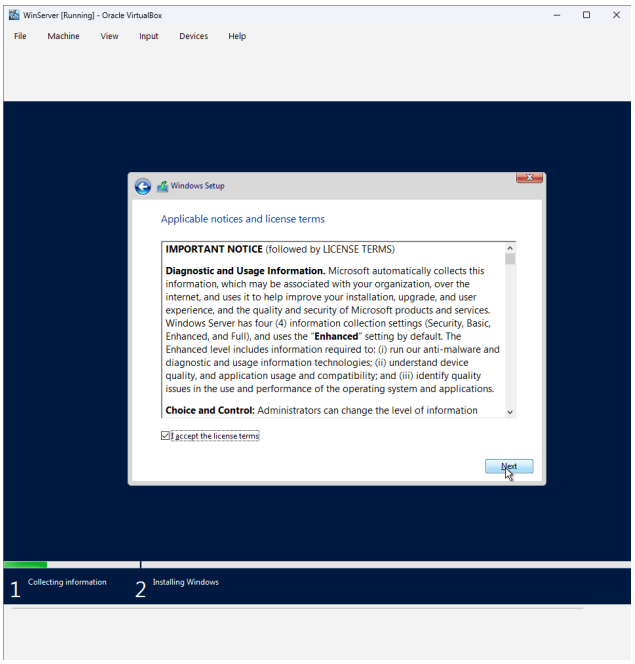


Figure 2.9: Installation Screen 3.

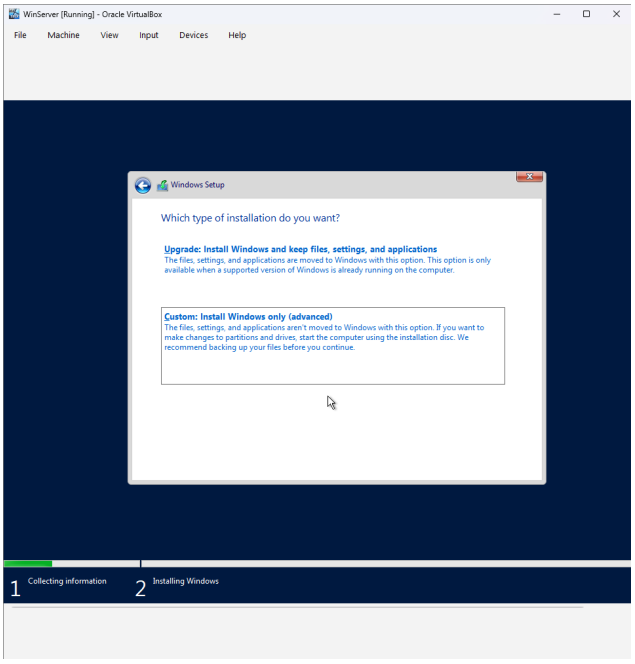


Figure 2.10: Installation Screen 4.

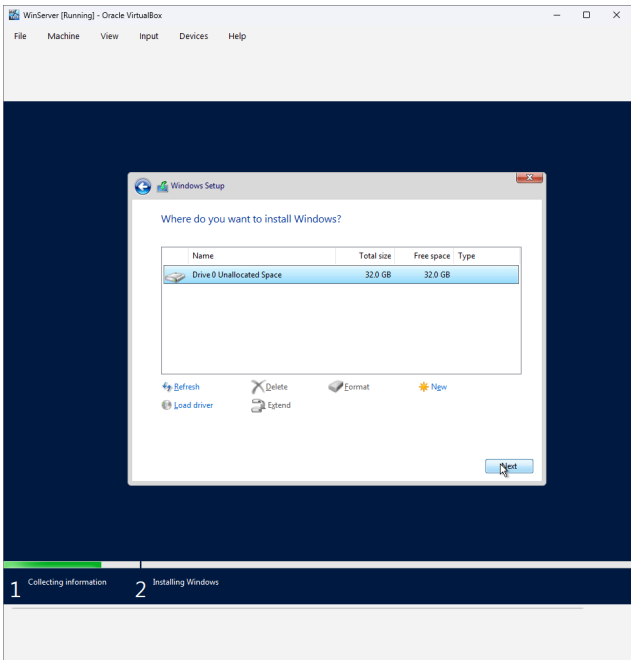


Figure 2.11: Installation Screen 5.

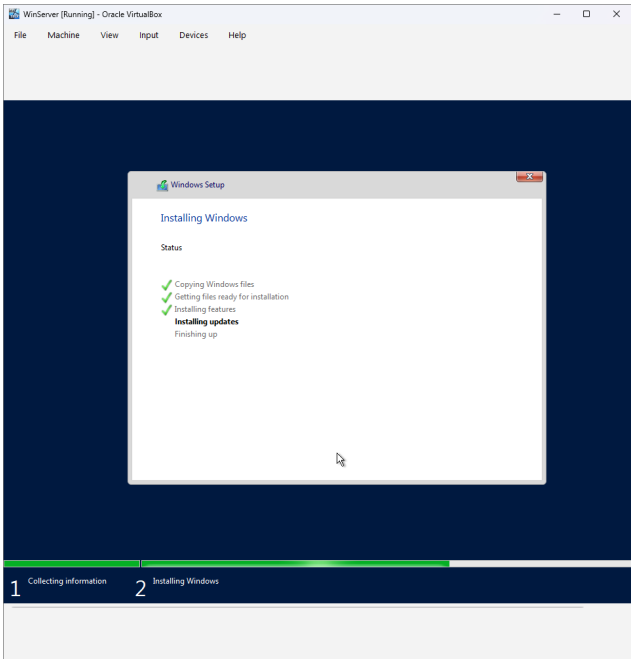


Figure 2.12: Installation Screen 6.

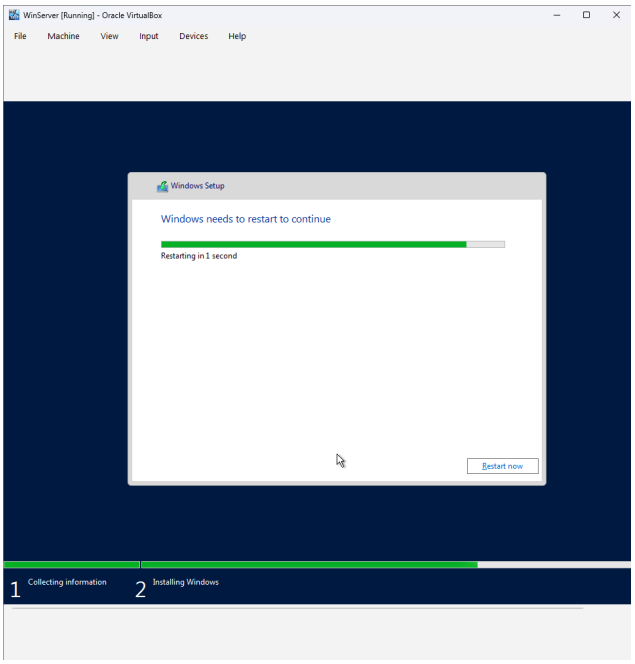


Figure 2.13: Installation Screen 7.

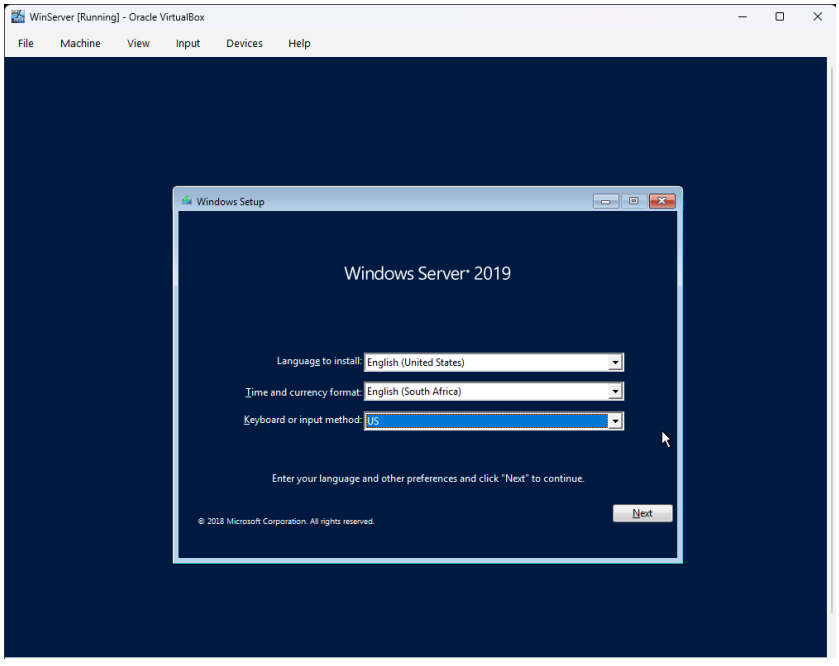


Figure 2.14: Installation Screen 8.

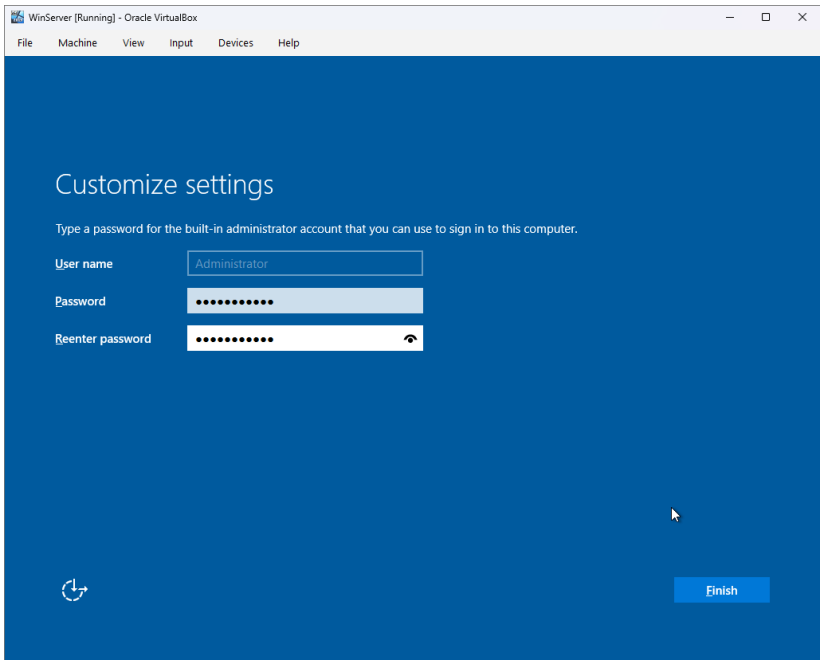


Figure 2.15: Windows Server Account Set-Up.

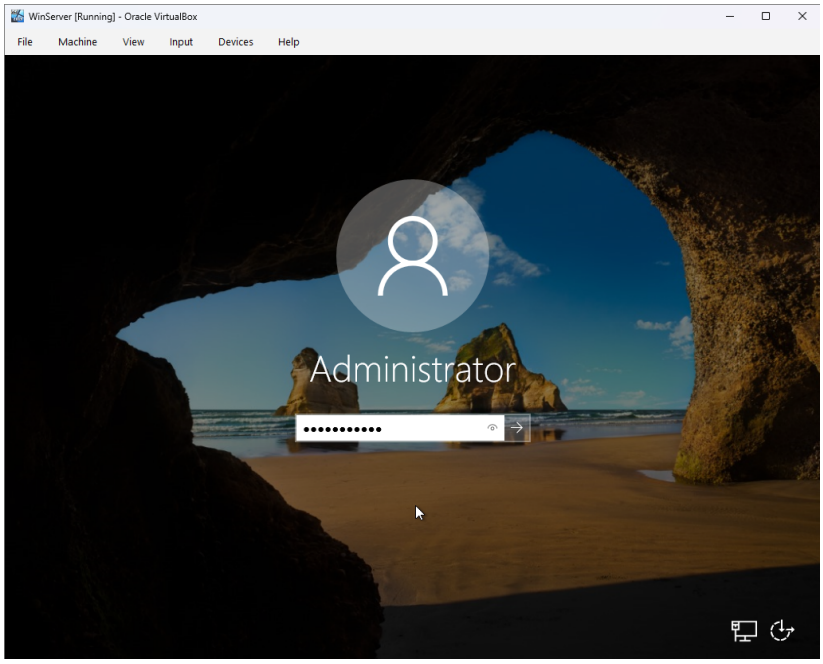


Figure 2.16: Windows Server Log-In Screen.

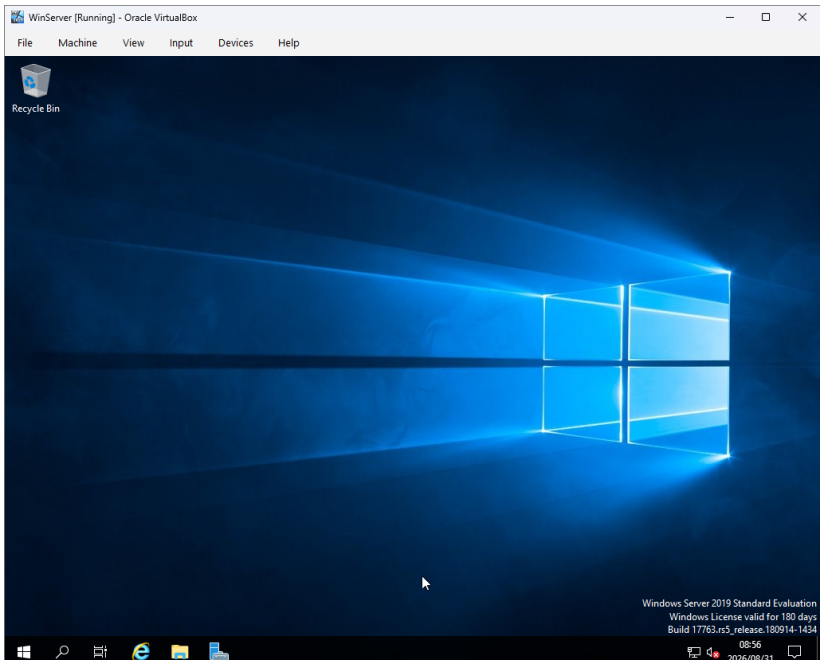


Figure 2.17: Windows Server Loaded and installed.

2.4 Testing

Testing was conducted to verify that Windows Server 2019 had been installed successfully and that the initial server configuration was functioning as expected. The tests focused on operating system availability, server identification, network configuration, and basic network connectivity.

2.4.1 Operating System Verification

The server was restarted to confirm that Windows Server 2019 could boot successfully after installation. The administrator was able to log in, and the Server Manager dashboard was accessible. This confirmed that the operating system was operational and ready for further configuration.

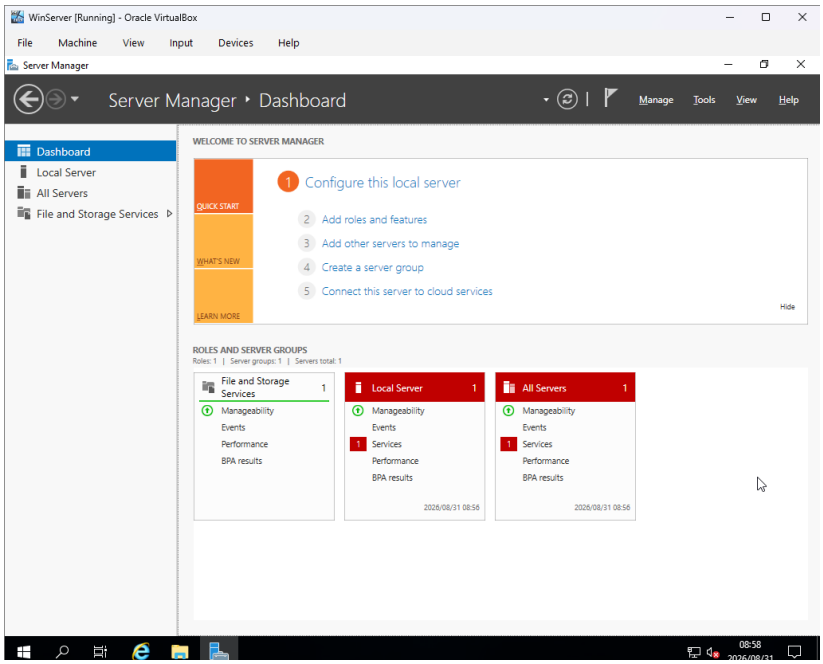


Figure 2.18: Server Manager dashboard following the installation of Windows Server 2019.

The dashboard provided access to the server's management tools and the interface used to install and configure additional server roles and features.

Further testing should verify the server hostname, IP configuration, and network connectivity. The following commands can be used to perform these checks:

- `hostname` – displays the configured computer name.
- `ipconfig /all` – displays the network adapter configuration.
- `ping <gateway-IP>` – tests connectivity to the configured network gateway, where applicable.

The outputs should be recorded and included as supporting evidence. Any failed tests should be investigated and resolved before proceeding with the configuration of additional server roles.

More comprehensive testing of Active Directory Domain Services, DNS, DHCP, Remote Access, and Windows Certificate Services is addressed in the relevant chapters of this report.

2.5 Chapter Summary

This chapter described the preparation of the virtual machine, installation of Windows Server 2019, and completion of the initial server configuration. The process established the operating system environment required for the deployment of additional server roles and network services.

Initial verification procedures were also described to assess operating system availability, server identification, network configuration, and connectivity. These steps establish the foundation for the remaining stages of the project.

The next chapter discusses the configuration and testing of Active Directory Domain Services (AD DS), which provides centralised management of users, computers, and authentication within the network environment.

3. Group Policy setup and configuration for Windows Server 2019

3.1 Introduction

Group Policy is a feature of Microsoft Windows that provides centralized management and configuration of operating systems, applications, and users' settings in an Active Directory environment. It allows administrators to define security settings, software installation policies, and user interface configurations across multiple computers within a domain.

Well implemented Group Policy can reduce risk, improve security, and ensure consistency across the network. Group Policy Objects (GPOs) can be define for specific users, groups, or computers, and can be linked to organizational units (OUs) to apply settings based on the structure of the Active Directory.

This chapter will cover the setup and configuration of Group Policy in a Windows Server 2019 environment, including the creation of GPOs, linking them to organizational units (OUs), and testing their application.

3.2 How does Group Policy work?

Group Policy relies on the Active Directory infrastructure to apply settings to users and computers. When a user logs in or a computer starts up, the Group Policy engine retrieves the relevant GPOs from the domain controller and applies the settings defined within those GPOs. The order of precedence for applying GPOs is as follows:

1. Local Group Policy: Settings defined on the local computer.
2. Site-level GPOs: Settings defined for the site in which the computer resides.
3. Domain-level GPOs: Settings defined for the entire domain.
4. Organizational Unit (OU)-level GPOs: Settings defined for the specific OU in which the user or computer resides.

When multiple GPOs are applied, the settings from the highest precedence GPO will override those from lower precedence GPOs. This allows administrators to create a hierarchy of policies that can be tailored to specific users, groups, or computers.

Policies are refreshed periodically (every 90 minutes by default), and administrators can also force a Group Policy update using the ‘gpupdate‘ command. This ensures that any changes made to GPOs are applied immediately.

3.3 Assignment specified Group Policy configuration

In this scetion we will cover the specific Group Policy configuration required for the assignment. The policies required include disabling command prompt, restricting removable storage access, preventing software installation, and enforcing password complexity requirements. Each of these policies will be configured through the Group Policy Management Console (GPMC) and linked to the appropriate organizational units (OUs) within the Active Directory.

The sections that follows will provide step-by-step instructions for creating and configuring the necessary GPOs, as well as testing their application to ensure they are functioning as intended.

3.4 Organizational Unit creation and configuration

The workflow for creating and configuring Group Policy Objects (GPOs) involves several steps, including accessing the Group Policy Management Console (GPMC), creating a new GPO, configuring the desired settings, and linking the GPO to the appropriate organizational unit (OU). Therefore the OU structure must be defined in Active Directory before creating and linking GPOs.

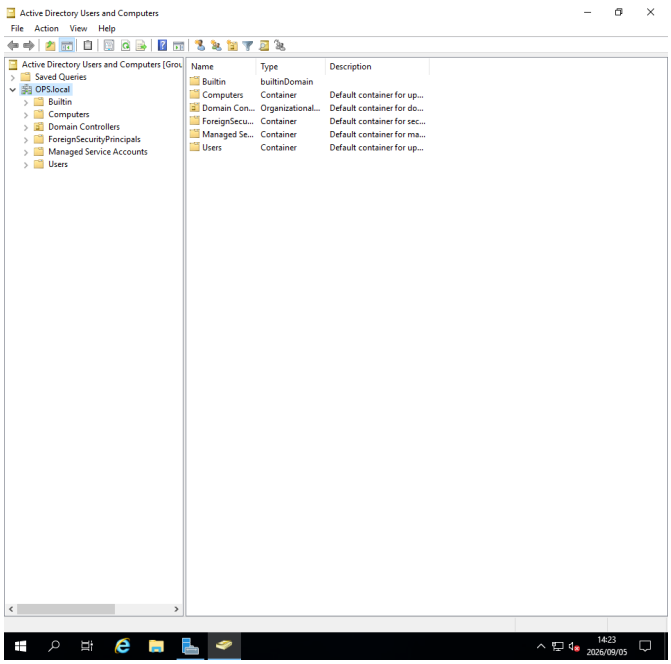


Figure 3.1: Active Directory Users and Computers.

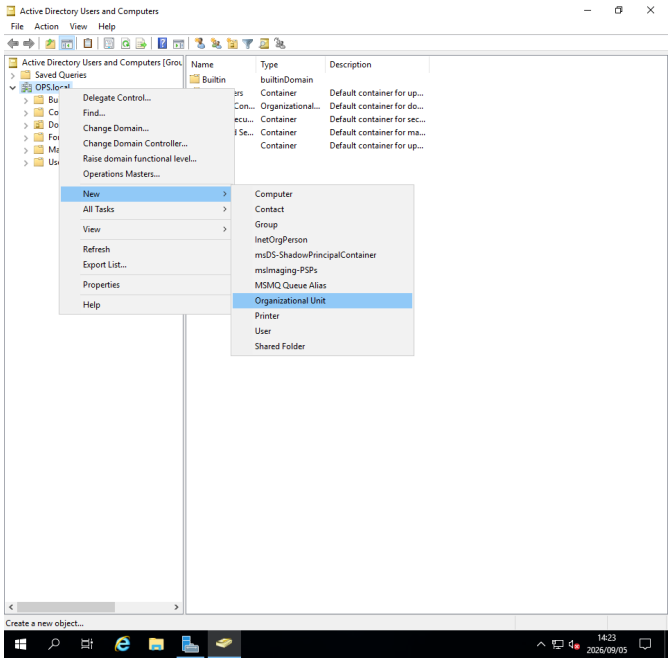


Figure 3.2: New OU object selected under the domain OPS.local.

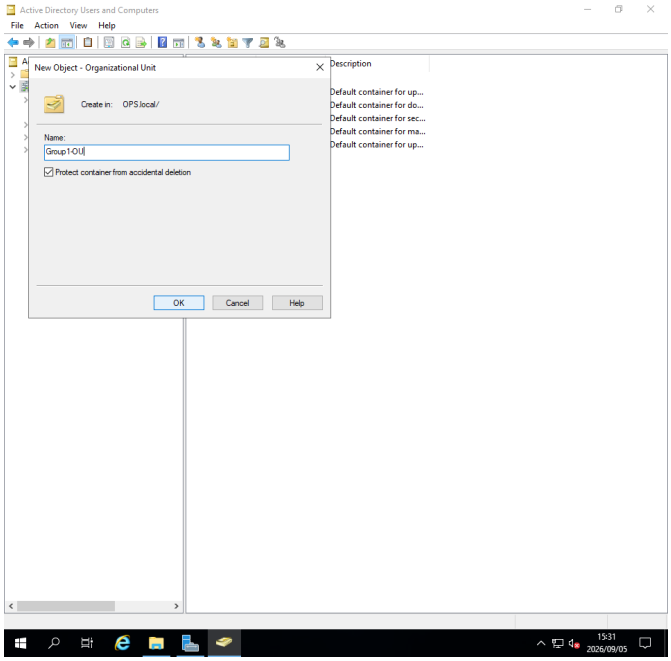


Figure 3.3: New OU object created and named Group1-OU.

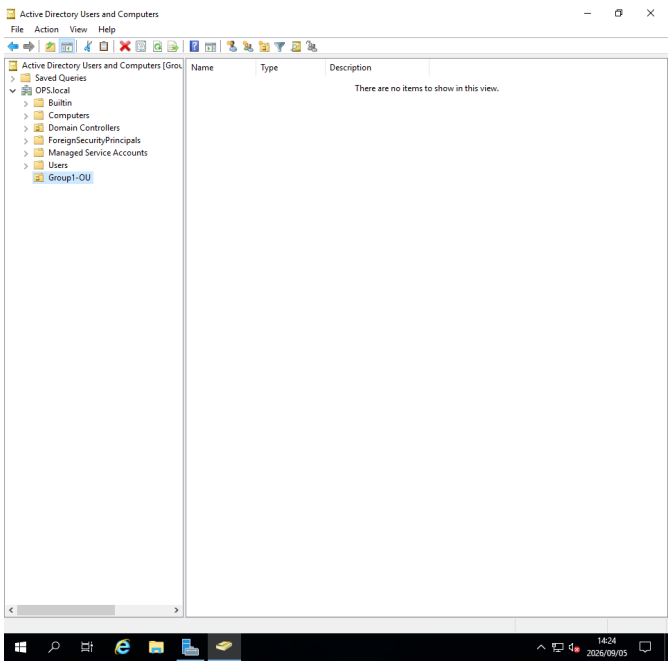


Figure 3.4: New OU object listed under the selected domain.

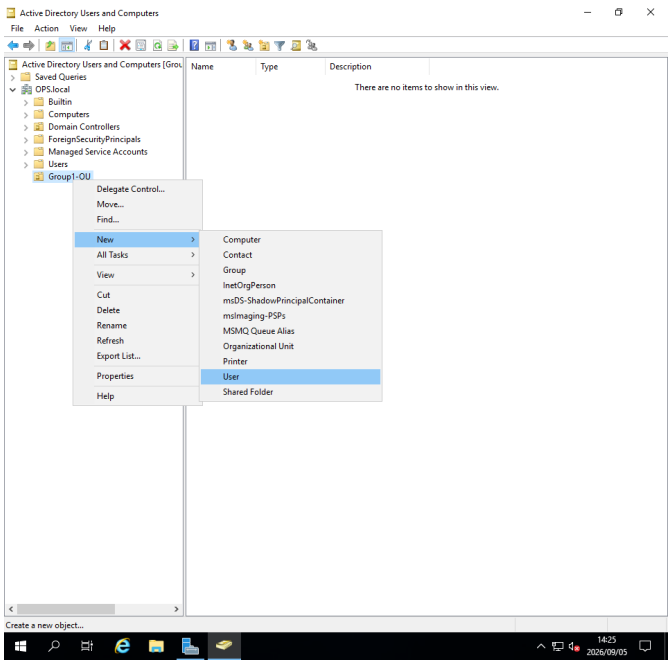


Figure 3.5: New User object selected under the Group1-OU.

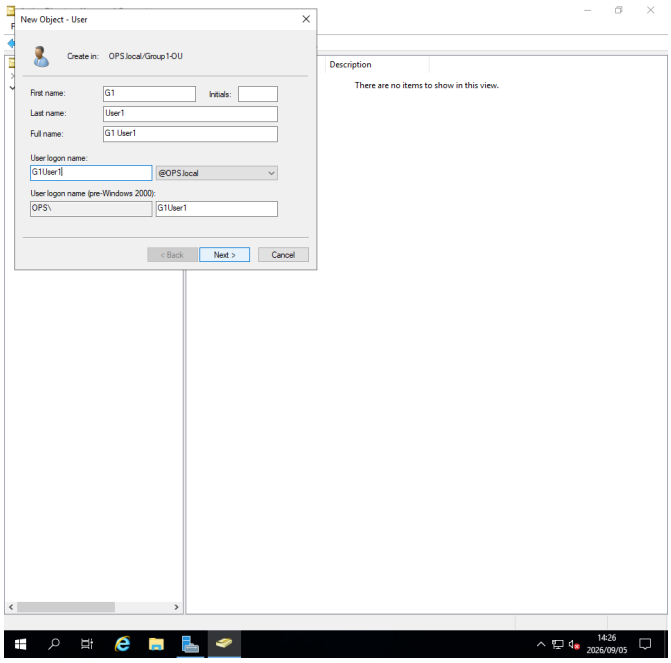


Figure 3.6: New User object configuration.

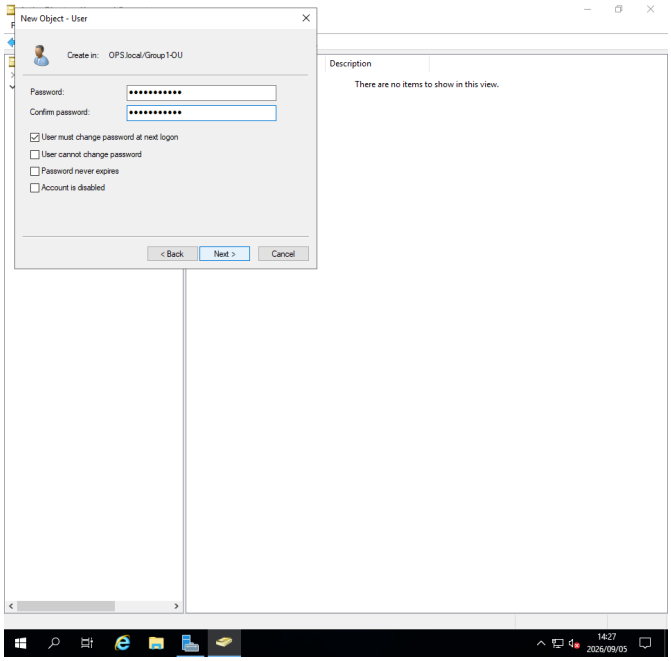


Figure 3.7: New User object password configuration.

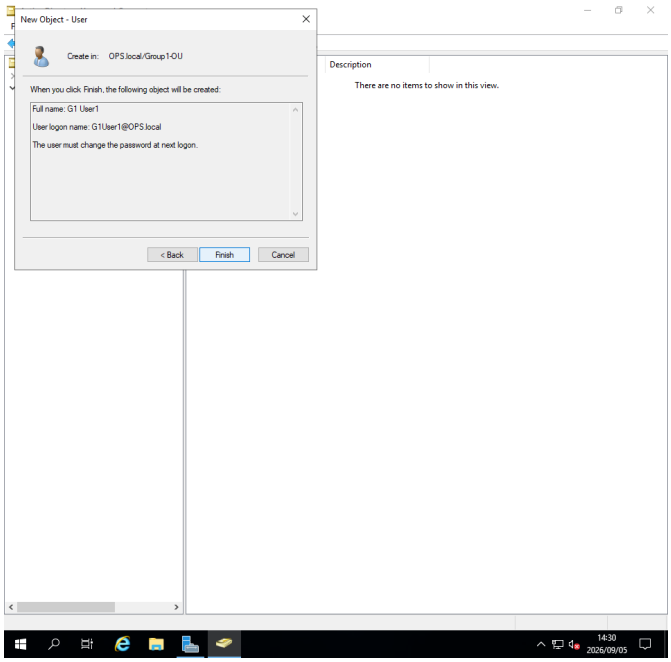


Figure 3.8: New User object summary.

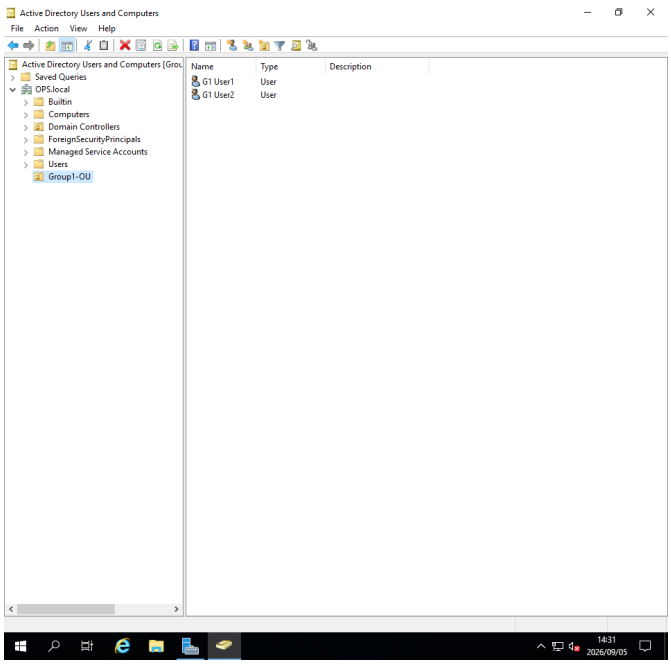


Figure 3.9: User object created and named. Here are two users listed.

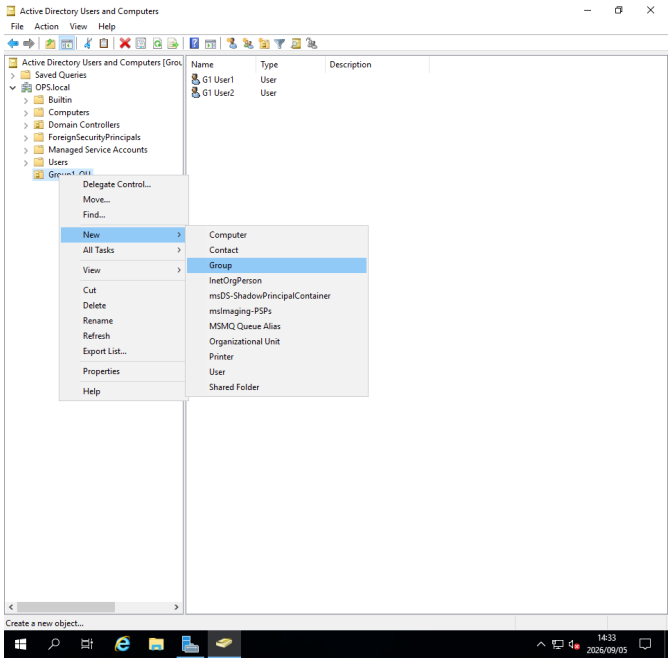


Figure 3.10: New Group Object selected under the Group1-OU. This allows the grouping of users.

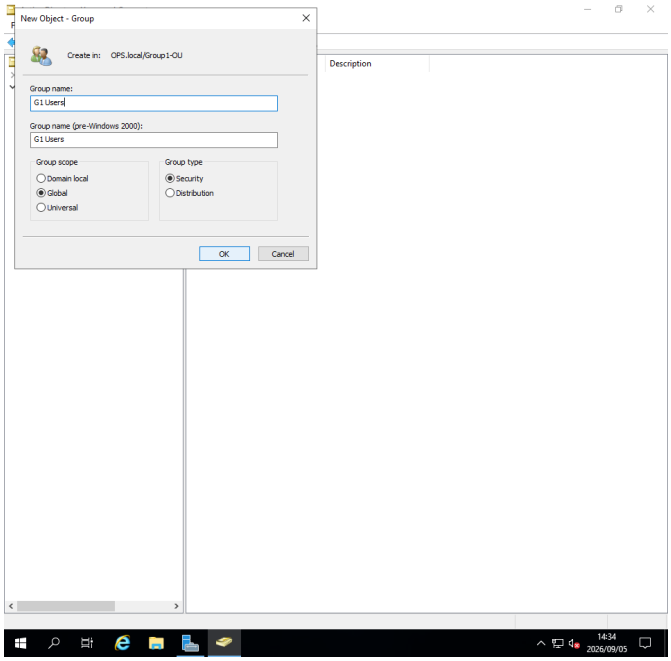


Figure 3.11: New Group object configuration. Group name, scope and type configured.

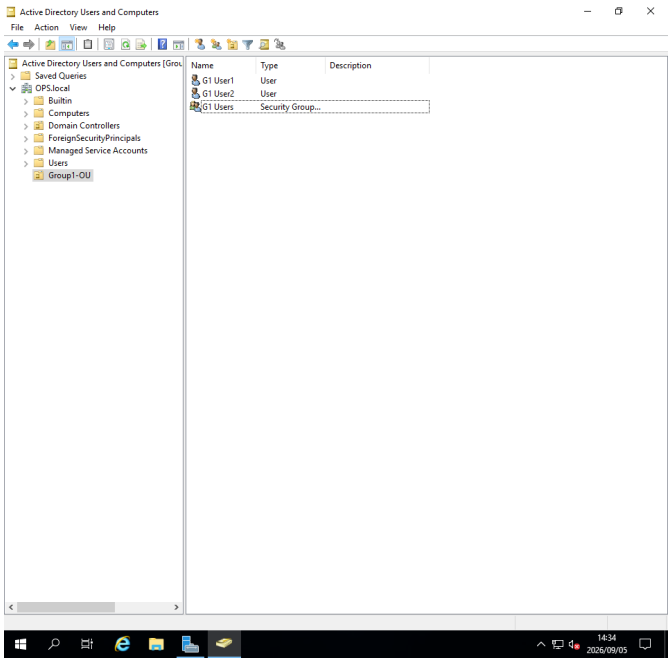


Figure 3.12: New Group object listed under the Group1-OU.

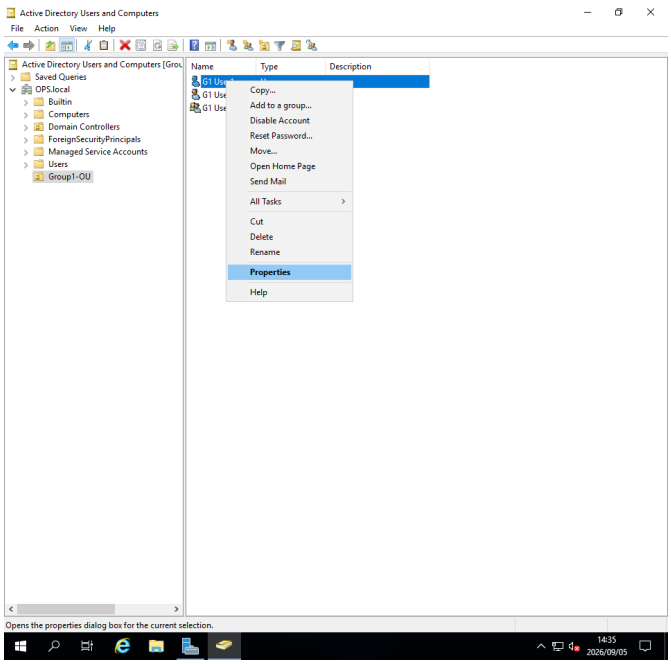


Figure 3.13: User1 showing membership configuration method 1.

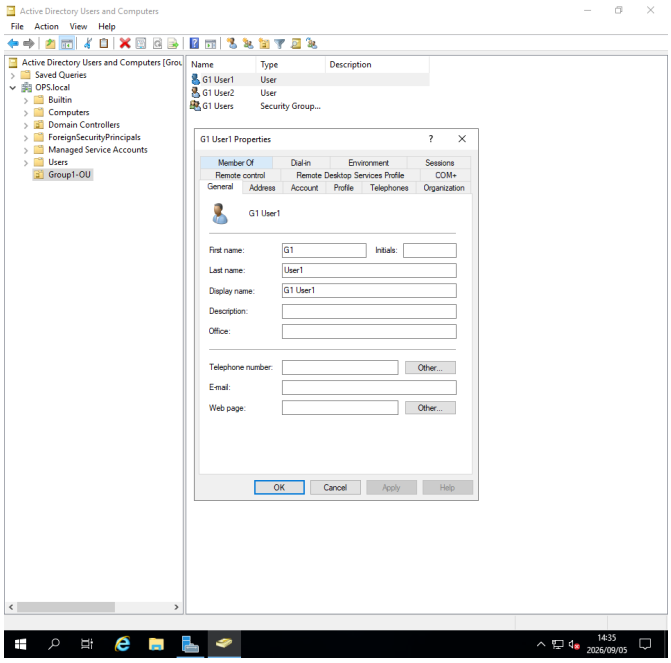


Figure 3.14: User1 membership configuration step 1.

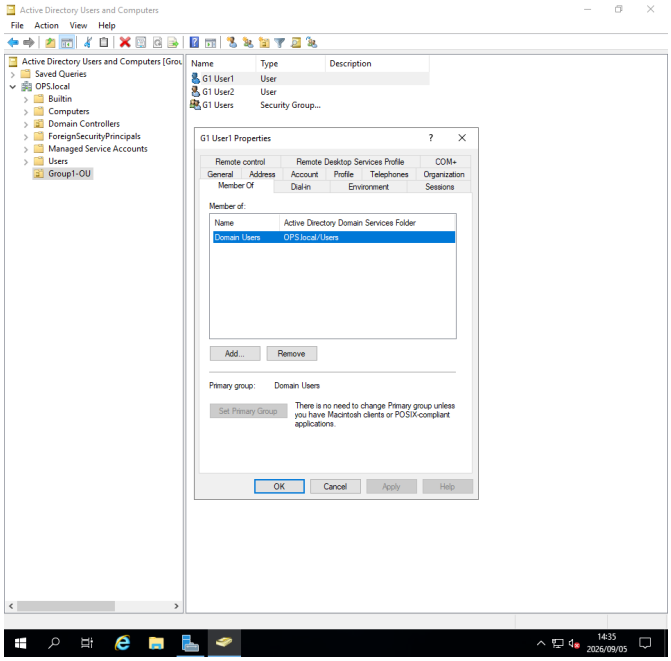


Figure 3.15: User1 membership configuration step 2.

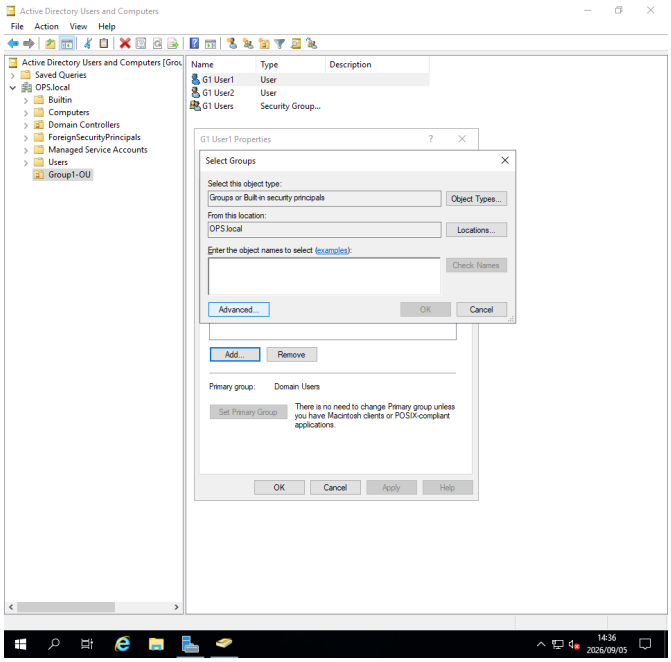


Figure 3.16: User1 membership configuration step 3.

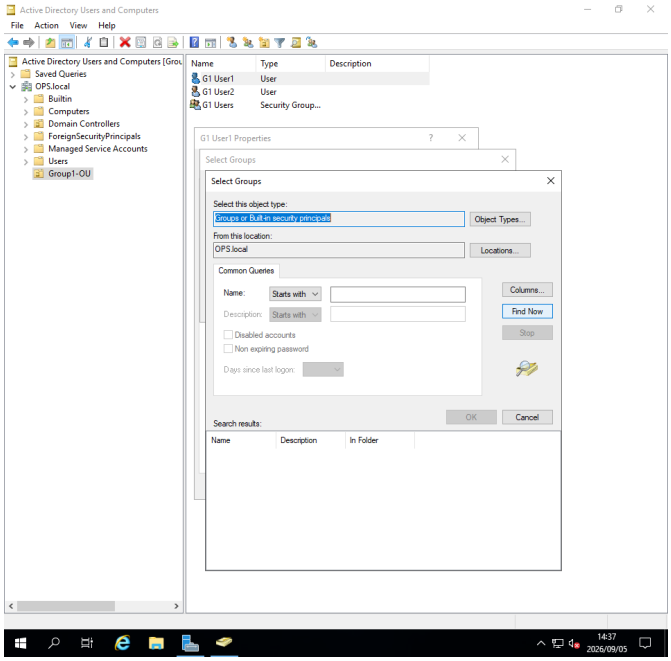


Figure 3.17: Group search window

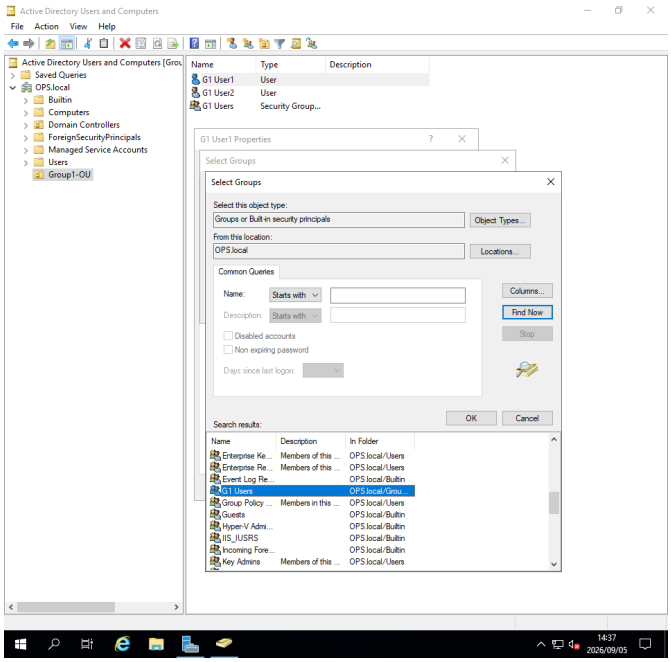


Figure 3.18: G1 Users Group find and selection.

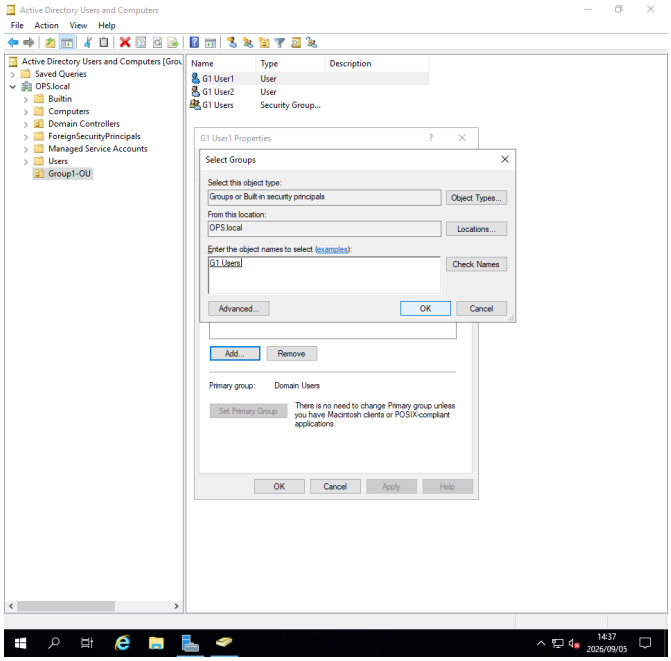


Figure 3.19: Group selection and addition window.

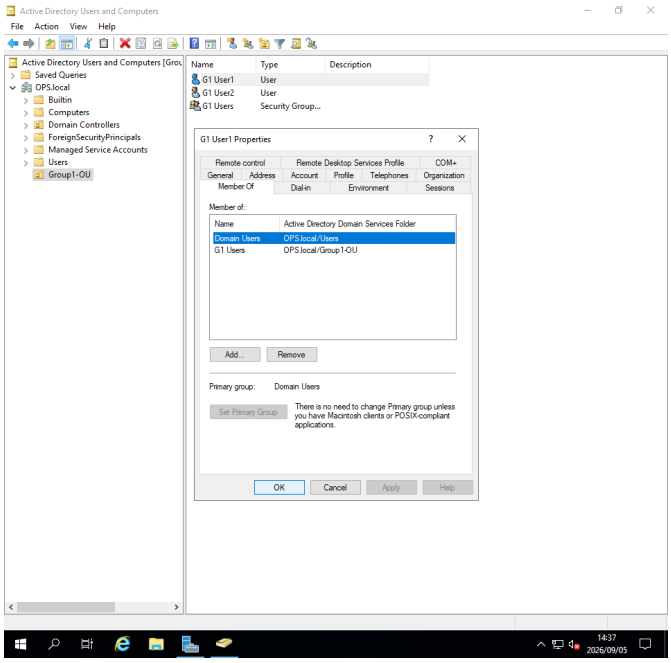


Figure 3.20: User1 membership tab.

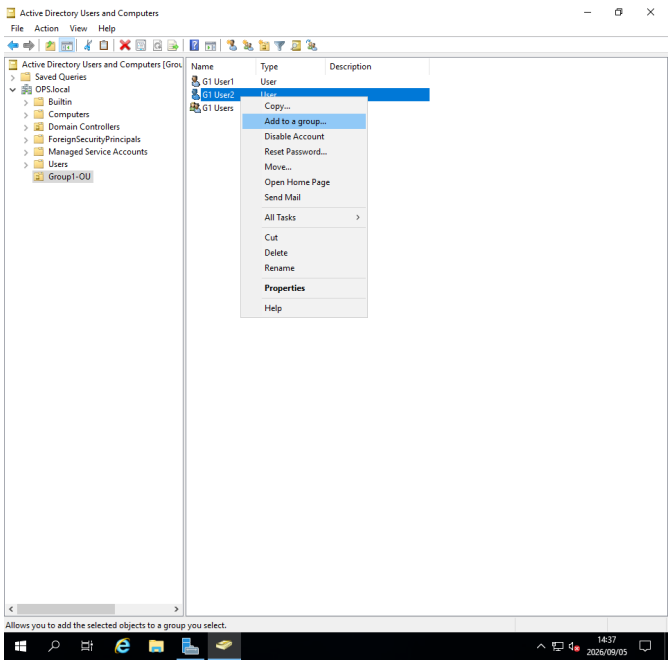


Figure 3.21: User2 showing membership configuration method 2. Add to a group selected.

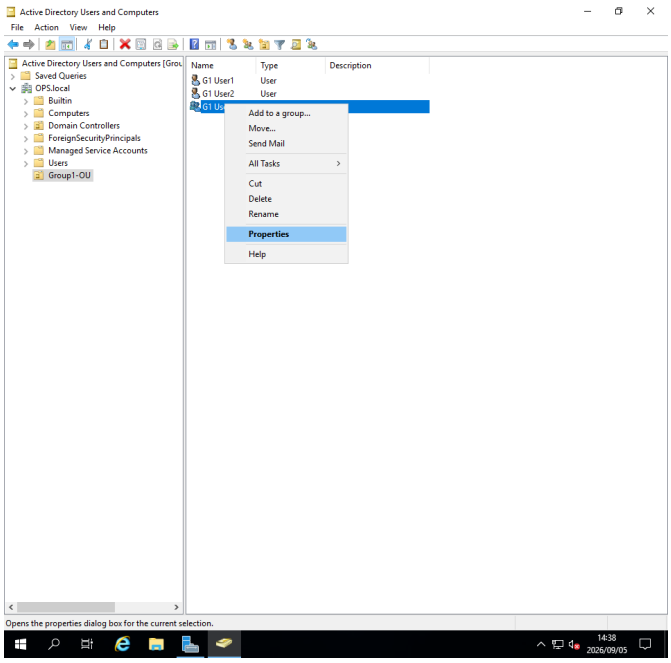


Figure 3.22: G1 Users Group properties selected to add Users to itself.

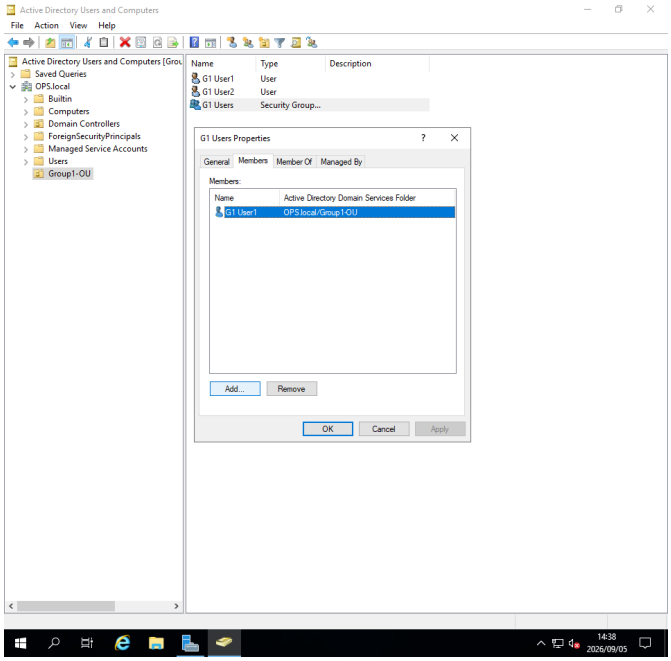


Figure 3.23: Properties tab with list of members for G1 Users Group.

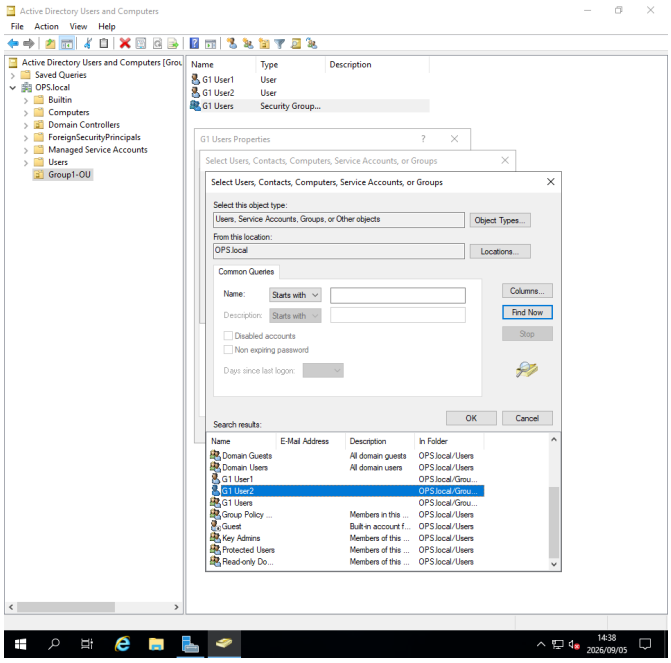


Figure 3.24: Users find and selection.

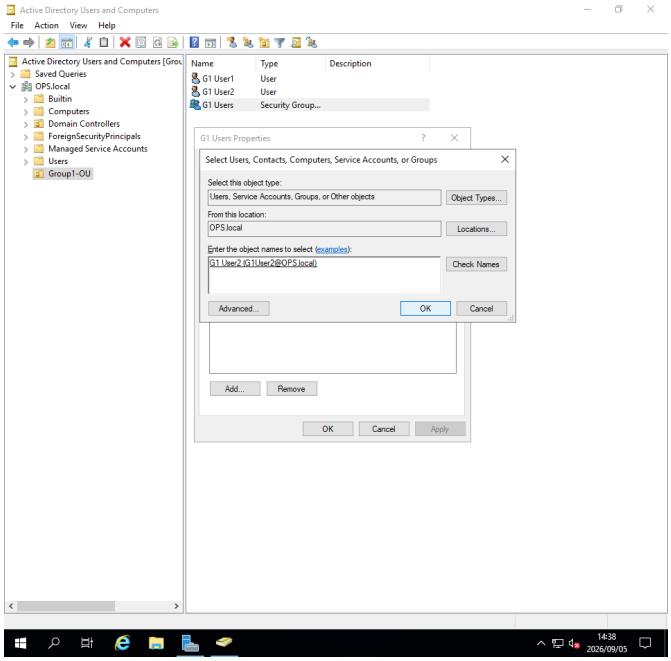


Figure 3.25: User2 selected and will be added to the group.

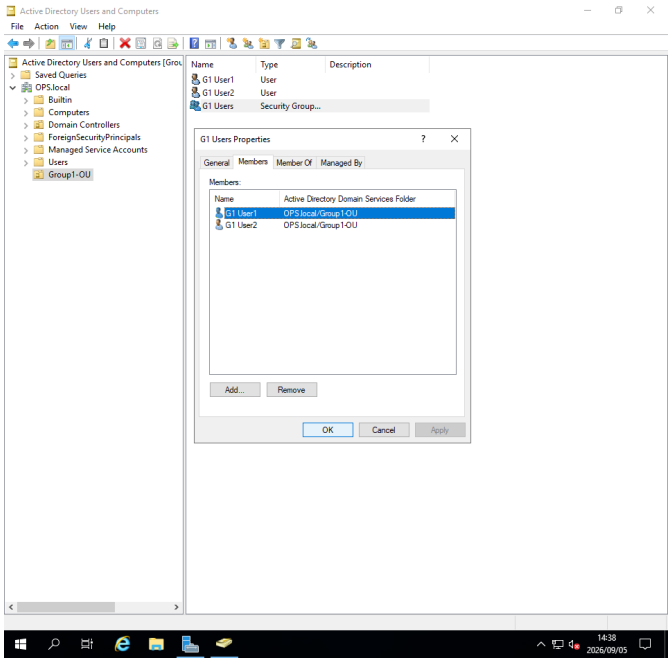


Figure 3.26: Properties tab with list of members for G1 Users Group. The list now includes User2.

3.5 Group Policy Object creation and configuration

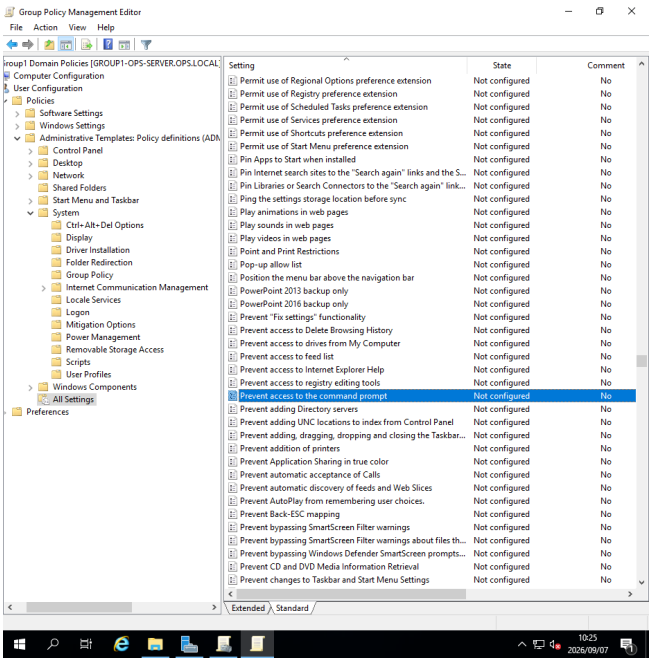


Figure 3.27: Group Policy Management Console. Prevent access to the Command Prompt selected.

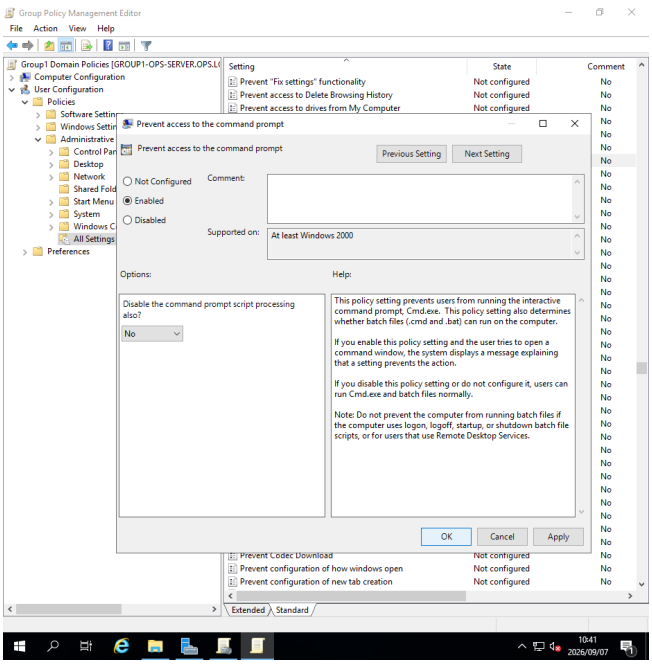


Figure 3.28: Policy settings for Prevent access to the Command Prompt. Enabled is selected.

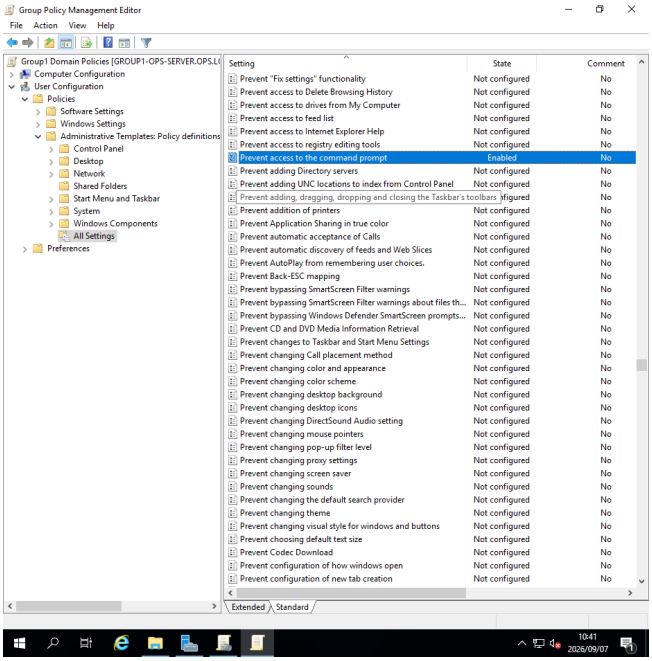


Figure 3.29: Policy settings for Prevent access to the Command Prompt shown as enabled.

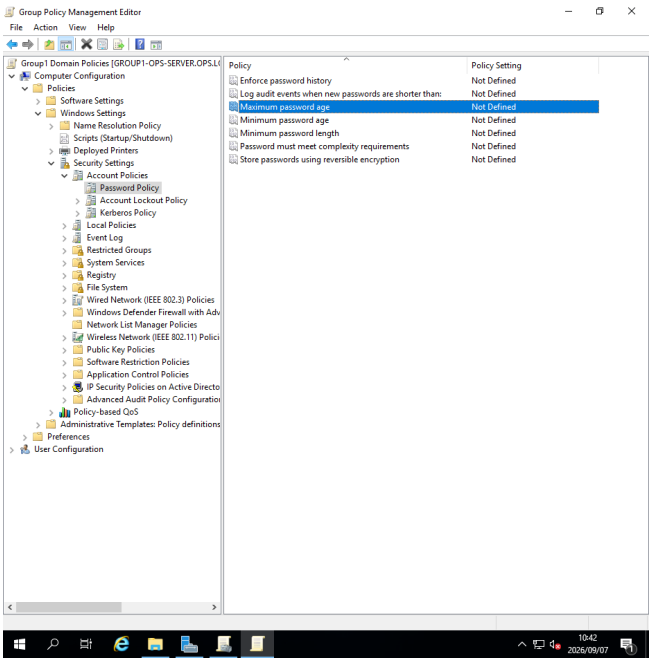


Figure 3.30: Password Policy settings. Maximum password age selected.

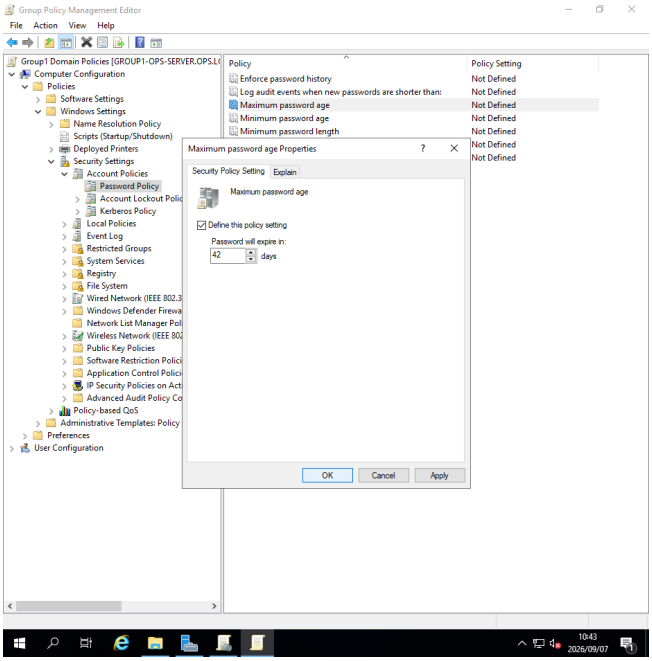


Figure 3.31: Maximum password age set to 42 days.

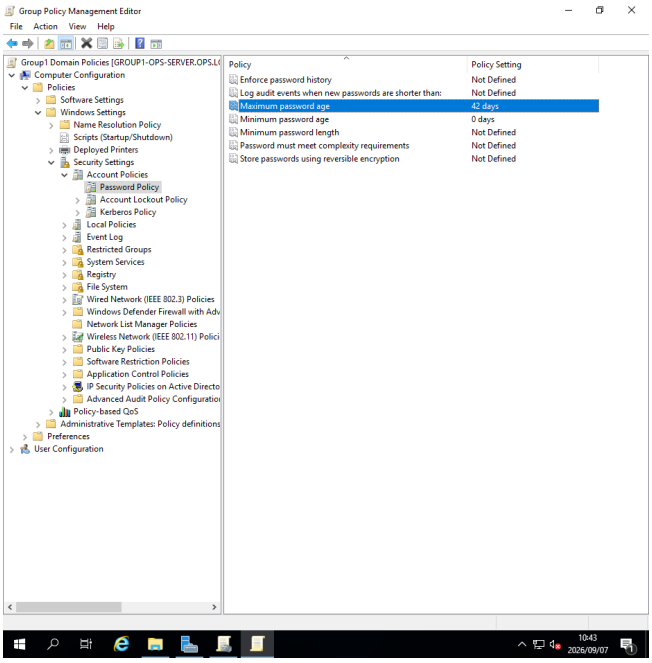


Figure 3.32: Maximum password age shown as 42 days.

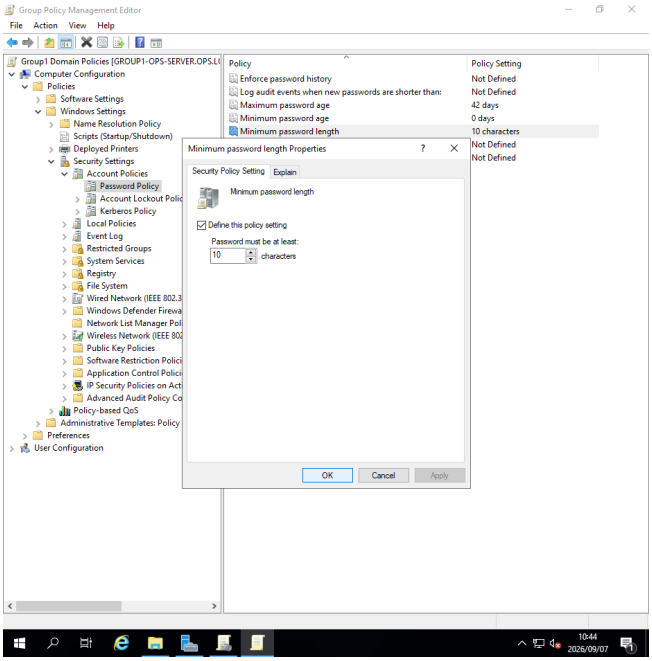


Figure 3.33: Minimum password length set to 10 characters.

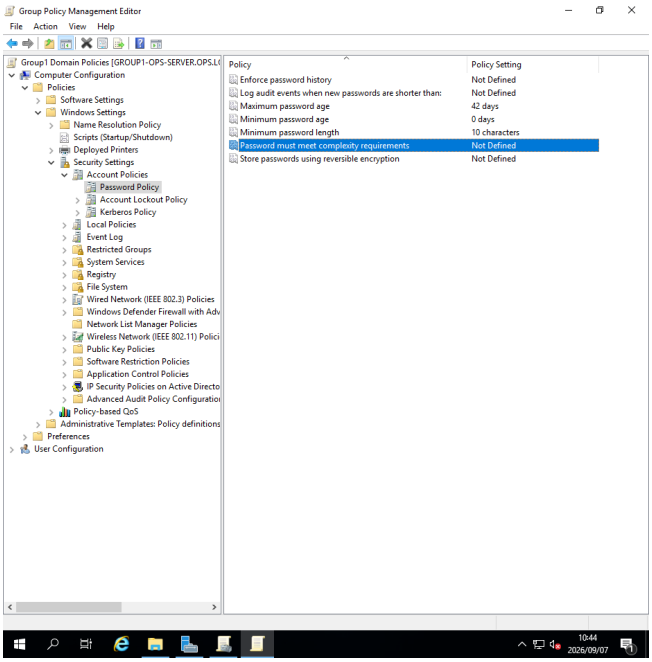


Figure 3.34: Minimum password length shown as 10 characters.

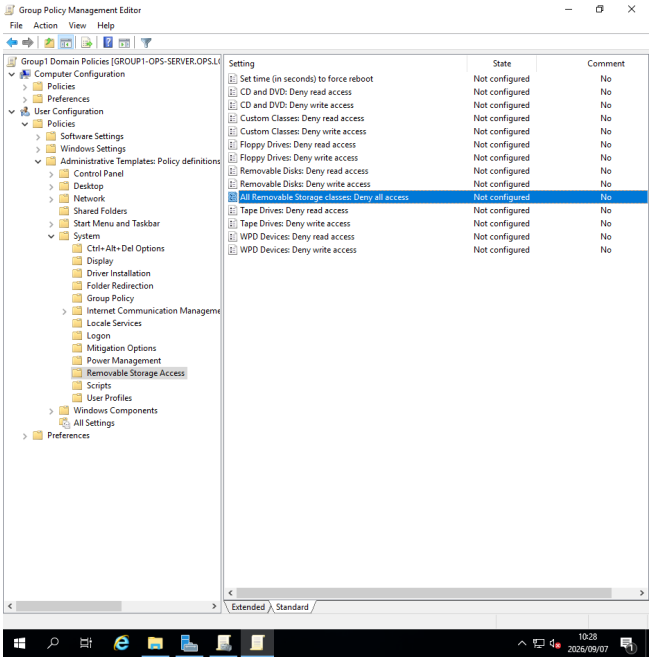


Figure 3.35: All removable storage classes: Deny all access selected.

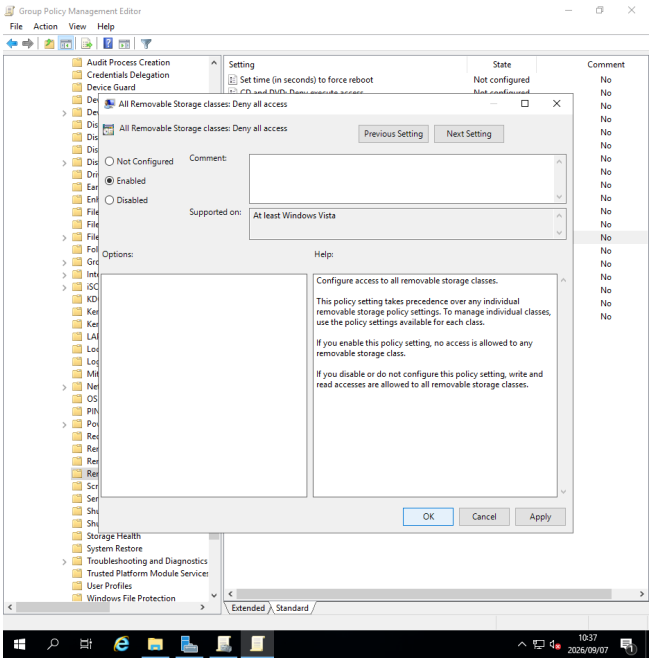


Figure 3.36: All removable storage classes: Deny all access Enabled.

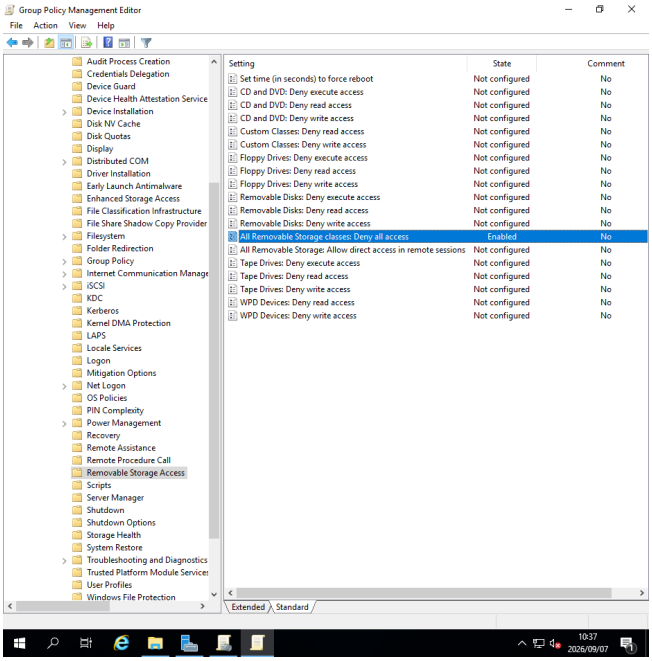


Figure 3.37: All removable storage classes: Deny all access shown as enabled.

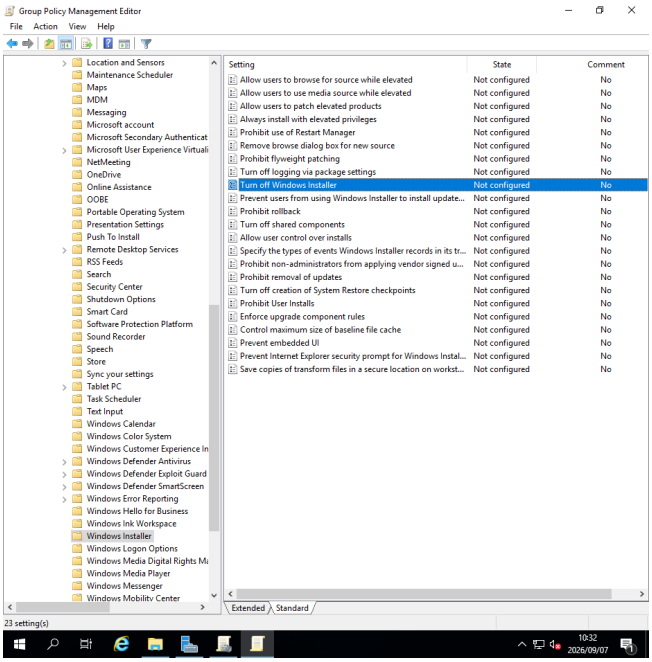


Figure 3.38: Turn off Windows Installer selected.

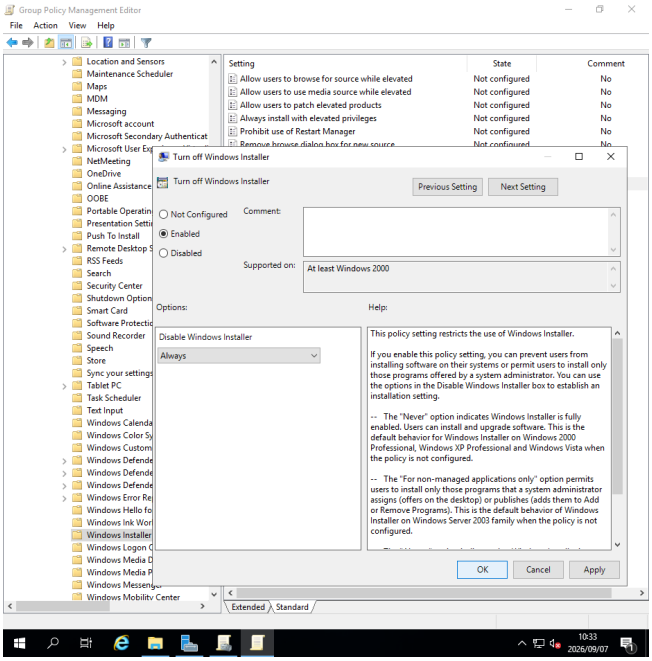


Figure 3.39: Turn off Windows Installer Enabled.

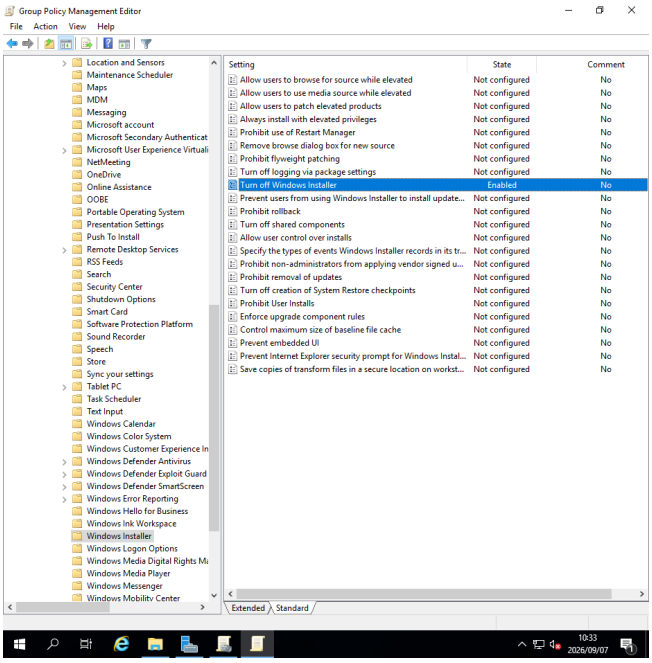


Figure 3.40: Turn off Windows Installer shown as enabled.

3.6 Testing

Testing is a critical step in the Group Policy configuration process to ensure that the policies are applied correctly and functioning as intended. After creating and linking the GPOs, administrators can use pre-deployment simulation tools to test the policies before applying them in a production environment. Additionally, administrators can

use the ‘gpresult‘ command to verify that the policies are being applied to the target users and computers. Furthermore, administrators can log in as a user affected by the GPOs to confirm that the restrictions and settings are enforced. The first and last methods of testing are quiet easily understood and therefore will not be covered in this report. The second method will be covered in the following section.

3.7 Chapter Summary

4. Conclusion

4.1 Summary

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4.2 Key Findings

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4.3 Future Work

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4.4 Final Remarks

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A. Appendix

A.1 Github Repository